

IBM battery 500

Get the in-depth dope on what makes up the Battery 500 project on this IBM site. It has videos and detailed diagrams. <http://dgit.in/O8C9bM>



Google Glass Demo

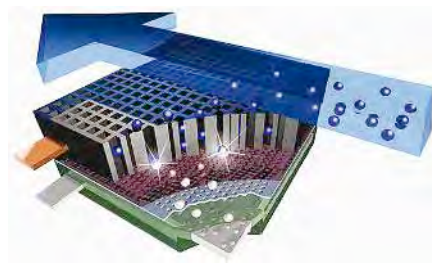
At Google I/O 2012, the Project Glass team took product demo presentation to a new level and even you can explore it. <http://dgit.in/MLYzfN>

From the Labs

heated. Though research is still under way, if Xu's "Graphene battery" is harnessing the thermal energy of the motion of ions to generate power, this may prove to be an essentially unlimited source of power – a huge breakthrough in self-powering technology.

IBM's Battery 500 – Batteries that breathe

Currently, electric vehicles can cover a maximum distance of around 100 km on one charge. Imagine an electric car that can



Battery 500: basic structure

run for around 800 km on a single charge. Well, if things move in the right direction, soon we may actually.

IBM successfully demonstrated its airbatteries, which draw oxygen from the air and recharge by expelling oxides. The concept of relying on the atmospheric air makes the batteries lighter and easier to accommodate inside cars as compared to conventional batteries which become too heavy for long journeys. If successfully implemented, we'll be able to travel from say, Mumbai to Bhopal on a single charge. Also, our dependence on fossil fuels would drastically decline. No more refueling at the petrol station! What's even

better is that once in mass production, the prices of these cars are expected to be in the price range of petrol cars.

Silicon Battery

Next in our list of promising batteries are silicon-based batteries. Silicon-based anode materials are in the process of being researched and manufactured by 3M, a leading battery material supplier in the US. The Department of Energy has given the project its nod, with a grant of \$4.6 million to build more energy-efficient vehicles. 3M invested a matching amount in resources. 3M was recently granted a U.S. patent for a Silicon-Lithium anode that promises a 20 per cent increase in energy density when used in conventional cathodes. If these anodes are paired with 3M's proprietary high-energy cathode technology, then the alloy can produce 40 per cent more electrons as compared to similar sized components. This effort has the potential to have a global impact on battery technology in terms of innovation. But don't run to the store just yet though, as this technology is expected to roll out within the next two years.

Battery, but no battery

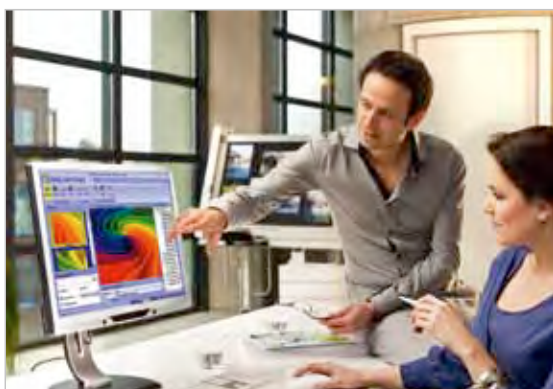
Again a work of MIT students, this device aims at providing a battery-free monitoring systems. Something that can be used in biomedical devices. The device can harness power from multiple sources such as natural light, body heat and vibrations in the environment. The latest work by a doctoral student, Saurav Bhandopadhyay has resulted in a chip that can harness power from all of the three power

sources mentioned above, optimizing power delivery. The major problem in this technology is having to switch between power sources. For example, body heat can generate 0.02 to 0.15 V whereas photovoltaic cells can generate 0.2 to 0.7 V and vibrations can produce up to 5 V. Lately, the approach to generating power in this way is to switch between power sources depending on the amount of power generated by each source. Co-ordinating the power sources to get a steady output though is tricky. Another approach is to extract power from all the sources and then switch rapidly between them.

Is "New Technology" actually required?

And then there's Powermat, a product that enables you to charge your devices wirelessly. To charge your device, you just need to attach a small receiver to the back of your phone and put it on your "Powermat". There's also a portable version that has a battery and you can carry it to charge your device when there's no power source around.

For cars, Qualcomm has developed WEVC (Wireless Electric Vehicle Charging) technology that can transfer energy wirelessly to your car batteries. Wireless transfer of electricity – or WiTricity as it's known – was demonstrated in 2007. Many successful commercial applications have come up in the market since then. Looking at the current trends in technology, even today we can overcome the problem of feeding our energy guzzlers. Guess we'll have to wait and watch which technology rises and shines and which ends up in the dust.



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